The 246 keV LZ nuclear recoil event and the LHC

October 2026

Caveat: I am absolutely not an expert, most of the material here I researched for the first time for this talk, likely made errors. Tell me if you think I misunderstood something.

But one question I will be raising will be this:
What if we have SUSY but with zero or almost zero ewkino mixing?

(This is me giving up a potential nice story climax
in order to make things easier to digest)

So what's the challenge for model builders?

Need to explain the event with high recoil energy but make sure you do not predict many low-recoil events (cause there arent any)

"Endothermic, boosted, q-dependent, exothermic"

Endothermic: we have an X1 N → X2 N process, with dm(X2,X1) approx 400 keV. Higgsinos might fit the bill.

Boosted: we are seeing boosted DM particles, not thermal ones
boost needs to be larger than LMC boost, what could boost them so much (Mukul might know)

Q-dependent: make the scattering process vastly q-dependent (pseudo-scalars as mediators?)

Exothermic: we have an X2 N → X1 N process
(what populates X2? Mukul might know)

So what's the challenge for model builders?

Need to explain the event with high recoil energy but make sure you do not predict many low-recoil events (cause there arent any)
[plus get the solar neutrino flux right]

For discussions re LC interplay, allow me to very broadly group into three categories:

A. electroweak multiplets (e.g. higgsinos)
For the endothermic, exothermic explanations we need X2, X1 with small mass gaps: (SUSY) electroweak multiplets to the rescue

B. DM + mediators (such as pseudoscalars)
For e.g. DM singlets, production via mediators, some mediators (e.g. pseudoscalars) would have an appropriate q-dependency

C. exotics
For e.g. the boosted DM explanation: hic sunt dracones, anything goes

A. Electroweak multiplets, e.g. higgsinos

SUSY-type model with (almost purely) higgsino DM models.

LHC implication: (most likely) higgsino searches

Current status: most interesting! Allow me to zoom in on this later

B. DM plus mediators

DM particles in the 100 – 1000 GeV range, mediators in the 500 – 5000 GeV range. Mediator could be axion-like, pseudoscalar (interaction needs strong q-dependence)

LHC implication: typical DM searches plus typical mediator searches, like resonance searches

Current LHC status: I am not aware of significant excesses in the resonance searches (but in SModelS we dont yet have much meta statistics on this)

C. boosted DM and other more exotics

LHC implications: could be anything, really. Monojets, sure.

Could be very low masses because relativistic.

Recap: higgsinos!

In SUSY we have two higgsino doublets (b/c we need two higgs doublets b/c holomorphism of superpotential b/c anomaly cancellation)

$$\tilde{H}_u = \begin{pmatrix} \tilde{H}_u^+ \\ \tilde{H}_u^0 \end{pmatrix} \qquad \tilde{H}_d = \begin{pmatrix} \tilde{H}_d^0 \\ \tilde{H}_d^- \end{pmatrix}$$

Generally, the higgsinos mix with the Winos and Binos to form the neutralinos and charginos

- The **neutral components** $\tilde{H}_u^0$ and $\tilde{H}_d^0$ mix with the **bino** ($\tilde{B}$) and **wino** ($\tilde{W}^0$) to form four **neutralino** mass eigenstates $\tilde{\chi}_i^0$ ($i = 1, \ldots, 4$).
- The **charged components** $\tilde{H}_u^+$ and $\tilde{H}_d^-$ mix with the **charged winos** ($\tilde{W}^\pm$) to form two **chargino** mass eigenstates $\tilde{\chi}_j^\pm$ ($j = 1, 2$).
- They are coupled in the superpotential via the μ**-term**: $W \supset \mu\, H_u \cdot H_d$.

Let's now assume that the LSP ("N1") is an (almost) pure higgsino. In this case, N1, N2, C1 are approximately mass degenerate.

For very large $M_{1,2}$, **one-loop electroweak corrections** can dominate:

$$\Delta m_{\text{1-loop}} \sim \tfrac{\alpha_2}{4\pi}\, m_W \sim \mathcal{O}(\text{few hundred MeV})$$

- $\tilde{\chi}_1^\pm - \tilde{\chi}_1^0$ **splitting**: For very large $M_{1,2}$, the tree-level $\sim m_W^2/M$ vanishes, but **one-loop EW corrections** ($\sim \tfrac{\alpha}{4\pi} m_W \sim 300$ MeV) take over, because electromagnetism breaks the $SU(2)$ symmetry relating charged and neutral Higgsinos.
- $\tilde{\chi}_2^0 - \tilde{\chi}_1^0$ **splitting**: The tree-level $\sim m_Z^2/M$ contribution **always dominates**, because in the pure Higgsino limit the two neutralinos form a **Dirac pair** with a protecting symmetry. This symmetry is only broken by gaugino mixing — not by gauge interactions — so one-loop EW corrections **cannot** generate an M-independent splitting.

higgsinos!

Summary: for a pure higgsino case we should
*) get an N2 that is basically mass degenerate with N1 (a 500 keV mass gap is basically mass degenerate here)
*) a C1 that is O(hundreds MeV) mass degenerate with N2, N1

N2 is then detector-stable, C1 becomes a long-lived particle (millimeters to meters of flightpath)

In typical SUSY at the LHC production of N1-N1 is negligible, N2-N1, C1-N1 and C1-C1 however are large.

N2-N1 should create “dark matter signatures” (monojets), C1-N1 and C1-C1 should create LLP and HSCP type signatures.

The other ewkinos (N3,N4,C2) would then be wino and bino-like, and at any other masses.

Biggest excesses we have in the SModelS database

```
nouc:~/git/protomodels [ww|+ ]> ./ptools/analyseDBDict.py -d historic_database_stats
/321.dict -a -n 30 --add_topos -e
[Analyzer:a_dbd] reading historic_database_stats/321.dict
[Analyzer:a_dbd] 61 CMS and 72 ATLAS results
# 1: Z=[-2.00,1.08,4.47] CMS-EXO-20-004: monojet
      - TRV1nunu,TChiZISRll,TChiISR,TRV1,...
# 2: Z=[-1.50,0.90,3.84] ATLAS-SUSY-2018-42: charged LLPs, dE/dx
      - THSCPM2,TDTM1F,TDTM2F,THSCPM1,...
# 3: Z=[-0.18,1.05,3.27] ATLAS-SUSY-2018-16-hino: 2 soft l + jets, EWK (hino)
      - TChiWWoff,TChiZoff,TChiWZoff
# 4: Z=[-2.60,0.16,3.22] ATLAS-SUSY-2018-16: 2 soft l + jets, EWK
      - TSlepSlep,TChiWZoff
# 5: Z=[-2.17,-0.05,2.92] CMS-SUS-21-007: 1 l + top- or W-tag
      - T1tttt,T5WW
# 6: Z=[-1.30,0.77,2.90] ATLAS-SUSY-2017-03: multi-l EWK
      - TChiWZ
# 7: Z=[-0.13,1.01,2.89] CMS-SUS-16-047: gamma + jets, high HT
      - T5gg,T6gg
# 8: Z=[-2.36,0.00,2.82] CMS-SUS-17-010: 2 l stop
      - T2ttoff,TChipChimSlepSnu,T2tt,T6bbWW
# 9: Z=[-2.56,0.31,2.65] CMS-SUS-13-019: >= 2 jets, MT2
      - T2tt,T2,T2ttoff,T1ttttoff,...
#10: Z=[-1.92,-0.28,2.61] CMS-SUS-16-049: All hadronic stop
      - T2tt,T2cc,T6bbWWoff,T2ttoff,...
#11: Z=[-2.34,0.01,2.59] ATLAS-SUSY-2019-09: 3 l, EWK
      - TChiWH,TChiWZ,TChiWZoff
#12: Z=[-1.73,-0.01,2.57] CMS-SUS-16-050: 0 l + top tag
      - T1tttt,T5tctc,T1ttttoff,T2tt
#13: Z=[-1.22,0.18,2.49] CMS-SUS-20-004: 2 h(b b), EWK
      - T5HH,TChiHH
#14: Z=[-0.65,0.58,2.46] ATLAS-EXOT-2018-06: monojet
      - TRV1nunu,TChiZISRll,TChiISR,TRV1,...
```

N2-N1?

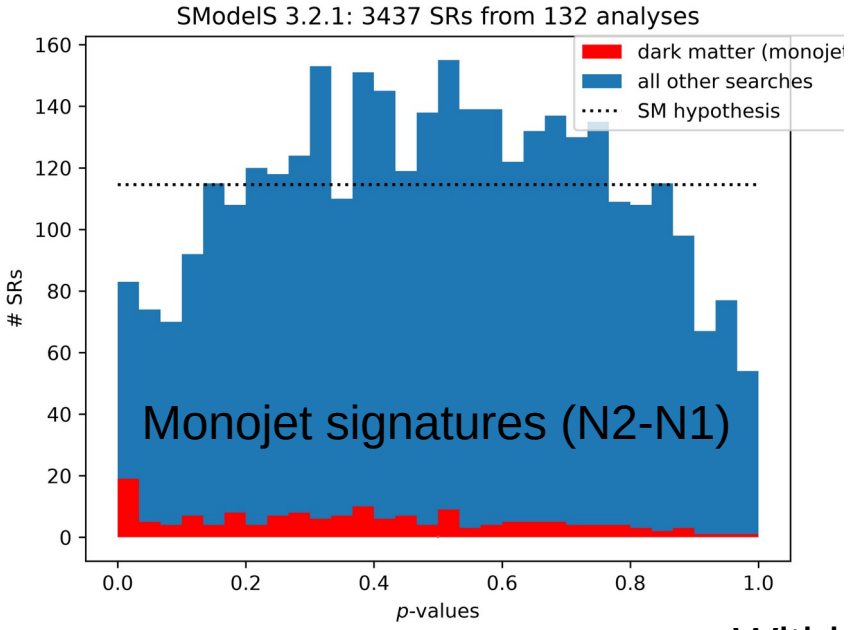
C1-N1?

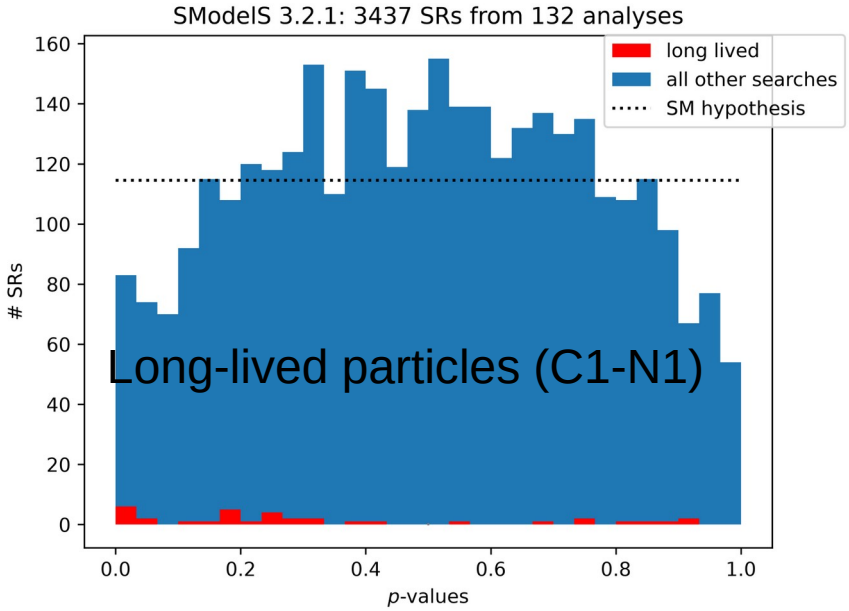
C*-N*?

Total number of analyses: 131 (or 163 if we count superseded)

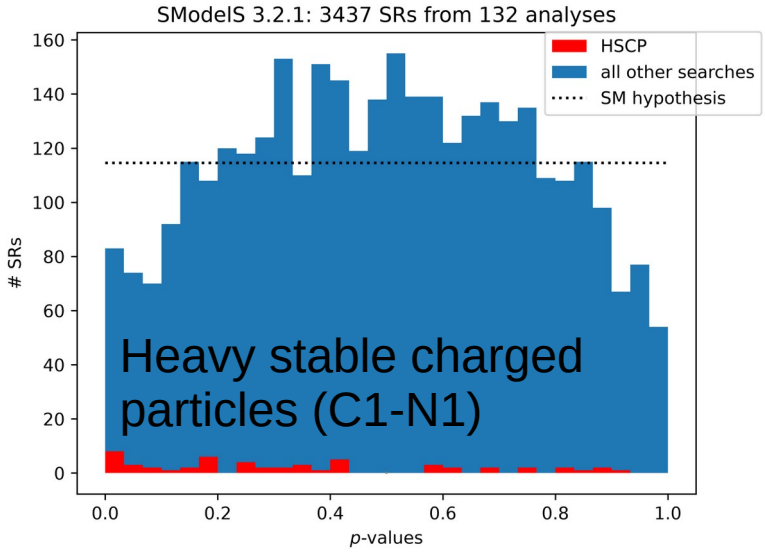
But: Many observations per analysis, complicated statistical models, complicated physics models

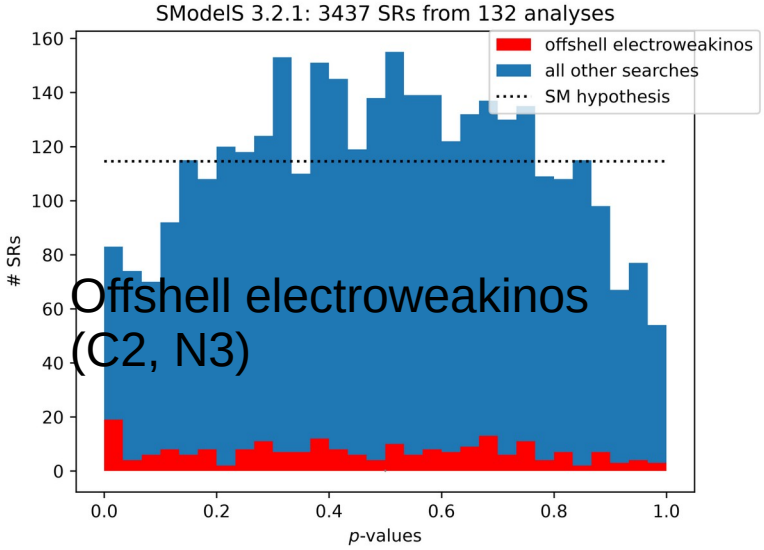
Biggest excesses we have in the SModelS database



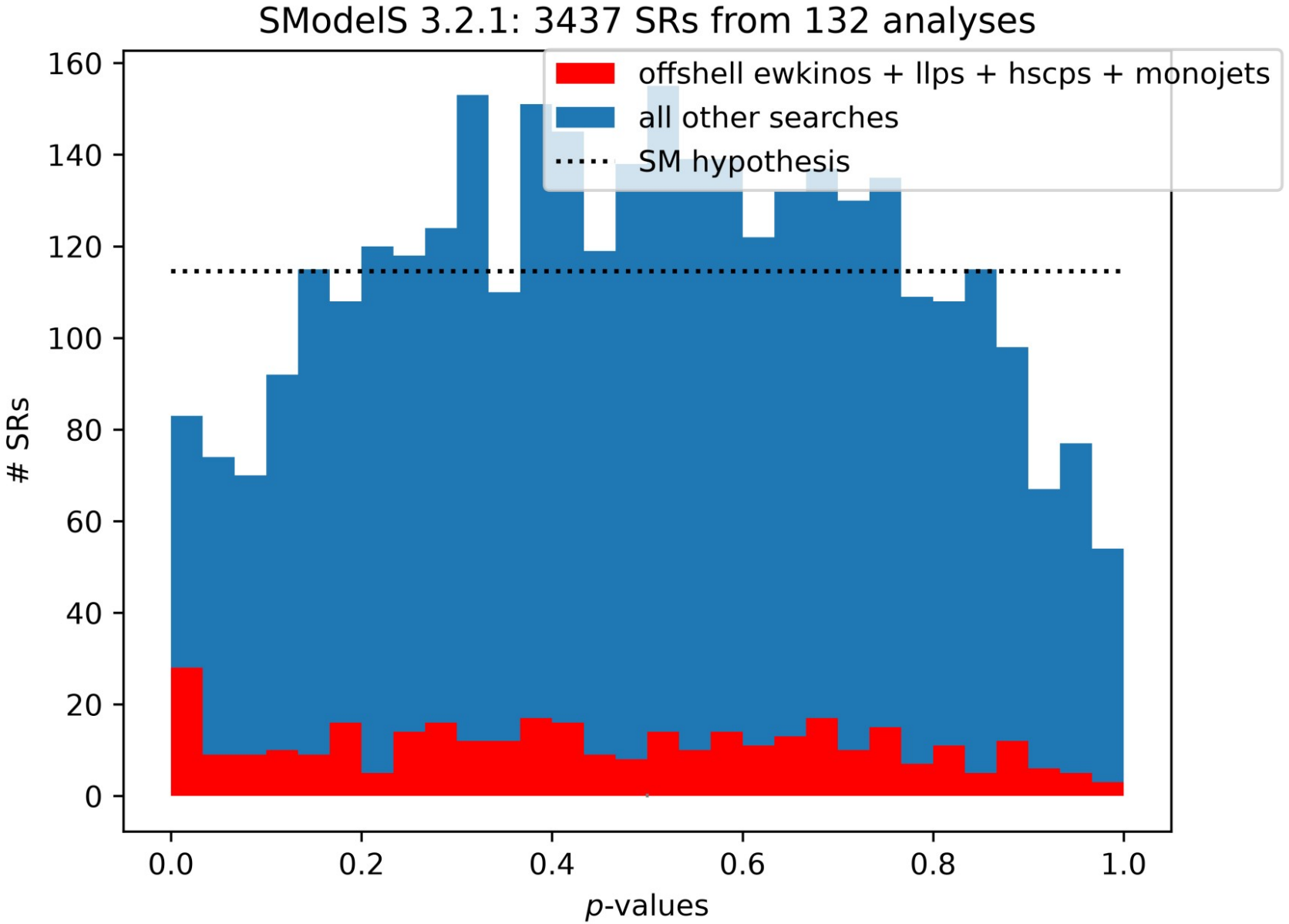


Within the MSSM: contradiction/tension between offshell electroweakinos and rest
Within e.g. the NMSSM: all fine



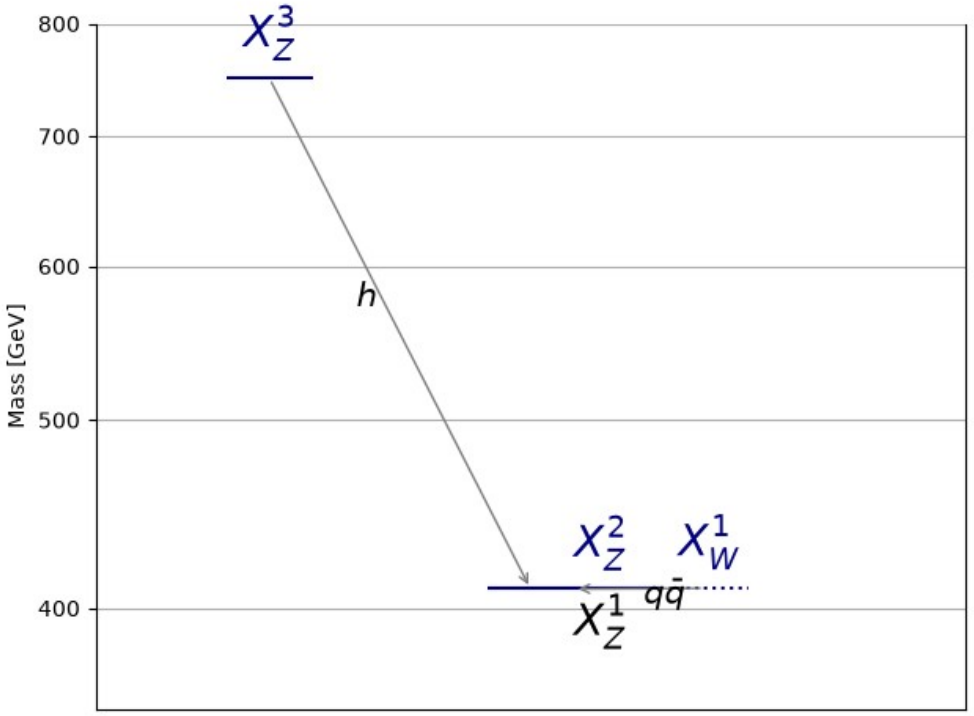


Biggest excesses we have in the SModelS database



For all the rest, we obtain an absurdly good agreement with the SM hypothesis

Wildly guessing, without our automated model building code running (because ran out of time):



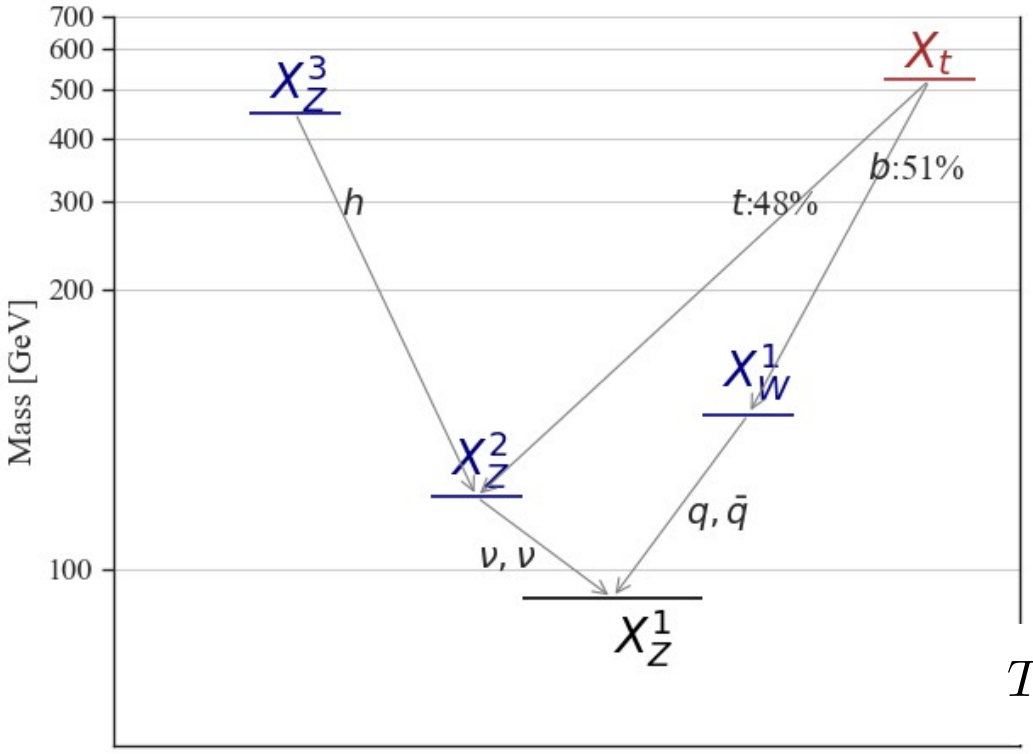
For pure higgsinos our excesses would want masses very roughly around 500 GeV. Here is a **handcrafted (NOT bestfit) model**

Passed our critic (so it is compatible with negative results from 150 analyses)#

K = 5 TL = 10, naively “around 3.2 sigma”

Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
ATLAS-EXOT-2018-06	em	EM10	TChiISR	413	359.00 +/- 10.00	2.4 σ	X^2_Z, X^1_Z
CMS-SUS-20-004	comb	-	TChiHH	0.2 fb	0.1 fb	1.5 σ	X^3_Z, X^2_Z
CMS-EXO-20-004	comb	-	TChiISR	3e+01 fb	2e+01 fb	1.4 σ	X^2_Z, X^1_Z
CMS-EXO-19-010	em	SR_nlay5	TDTM2F	6	3.70 +/- 1.24	0.8 σ	X^1_W, X^2_Z, X^1_Z
ATLAS-SUSY-2018-42	em	SR400_Low_long	TDTM2F	33	29.20 +/- 2.20	0.6 σ	X^1_W, X^2_Z, X^1_Z
ATLAS-SUSY-2016-06	em	SR_EW	TDTM2F	9	11.80 +/- 3.10	-0.7 σ	X^1_W, X^2_Z, X^1_Z

Not in Best Combo	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
ATLAS-SUSY-2018-19	em	SR_EW	TDTM2F	3	3.00 +/- 0.70	-0.2 σ	X^1_W, X^2_Z, X^1_Z



Thats what I had before

Didnt allow X2Z to be stable so my machinery invented a “trick” to make it invisible

(wino-like xsecs, hence lower masses)

$$T_L := 2 \ln \left(\frac{L^c_{\mathrm{BSM}}(\hat{\mu}) \cdot \pi_{\mathrm{BSM}}}{L^c_{\mathrm{SM}} \cdot \pi_{\mathrm{SM}}} \right) = 39.06 = 6.25^2$$

Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
CMS-EXO-20-004	comb	-	TChiISR, TChiZISRqq	2e+02 fb	1e+02 fb	4.6 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-EXOT-2018-06	em	EM10	TChiISR, TChiZISRqq	413	359.00 +/- 10.00	2.5 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-SUSY-2018-12	em	SRBTT	T2tt	67	46.00 +/- 7.00	2.1 σ	X_t, X^2_Z
CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.0 σ	X^3_Z, X^2_Z
ATLAS-SUSY-2018-08	em	SRSF140	T2tt	9	5.10 +/- 0.90	1.5 σ	X_t, X^2_Z
ATLAS-SUSY-2016-16	em	tN_med	T2tt	50	36.30 +/- 6.60	1.5 σ	X_t, X^2_Z
ATLAS-SUSY-2013-16	em	SRA4	T2tt	4	2.40 +/- 0.70	0.9 σ	X_t, X^2_Z
ATLAS-SUSY-2013-15	em	tNboost	T2tt	5	3.30 +/- 0.70	0.9 σ	X_t, X^2_Z
CMS-PAS-SUS-13-015	em	pTmiss350_Nb2	T2tt	15	8.60 +/- 7.10	0.8 σ	X_t, X^2_Z
CMS-SUS-16-050-agg	comb	-	T2tt	2 fb	2 fb	-0.2 σ	X_t, X^2_Z
CMS-SUS-13-011	em	LM150	T2tt	227	251.00 +/- 50.00	-0.5 σ	X_t, X^2_Z

Other pheno groups

Mini summary of MSSM:

A) wino/bino DM with chargino co-annihilation ($|M_1| \sim M_2 \lesssim \mu$)

relic DM density 100% fulfilled

$\Rightarrow$ fits well the excesses!

Never: $M_1 \sim M_2/2$ $\Rightarrow$ not GUT-based

D) higgsino DM: $m_{\tilde{\chi}_1^0} \sim m_{\tilde{\chi}_2^0} \sim m_{\tilde{\chi}_1^\pm} \sim \mu$ ($\mu \lesssim |M_1|, M_2$)

relic DM density as upper limit (full relic density: $m_{\tilde{\chi}_1^0} \sim 1$ TeV)

$\Rightarrow$ does not fit the excesses Possible: $M_1 \sim M_2/2$ $\Rightarrow$ can be GUT-based

No higgsino DM because they are assuming the MSSM! The downside of UV-complete theory driven approaches

S. Heinemeyer, SUSY2026

TODO:

Run our machinery with these types of scenarios in mind (started to take into account LLP/HSCP only this week)

TODO:

Where we have higgsinos, we should also have extra higgses. Need to add their search results to SModelS!

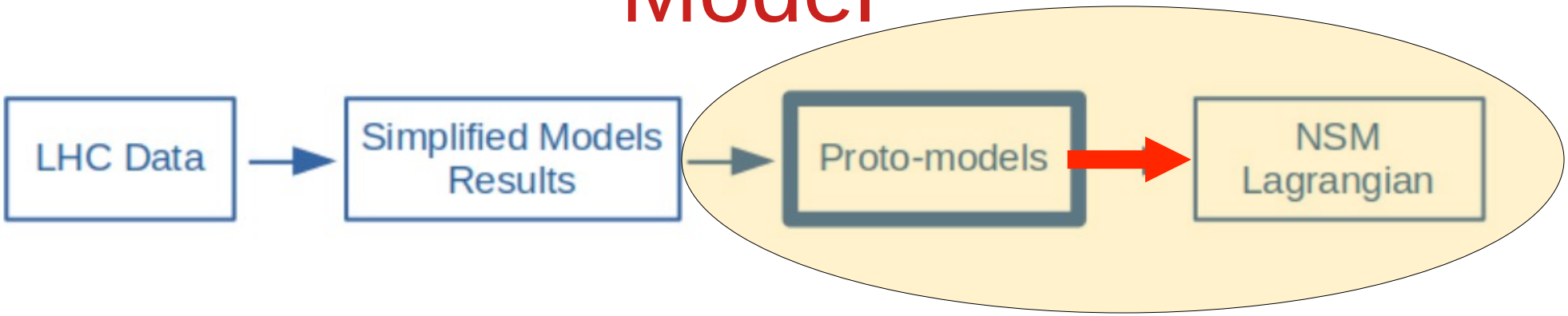
TODO:

If we want to explain the soft dileptons at dm approx 20 GeV, we cannot have pure higgsinos anymore.
Is there a sweet spot? NMSSM singlino? Something not thought of?

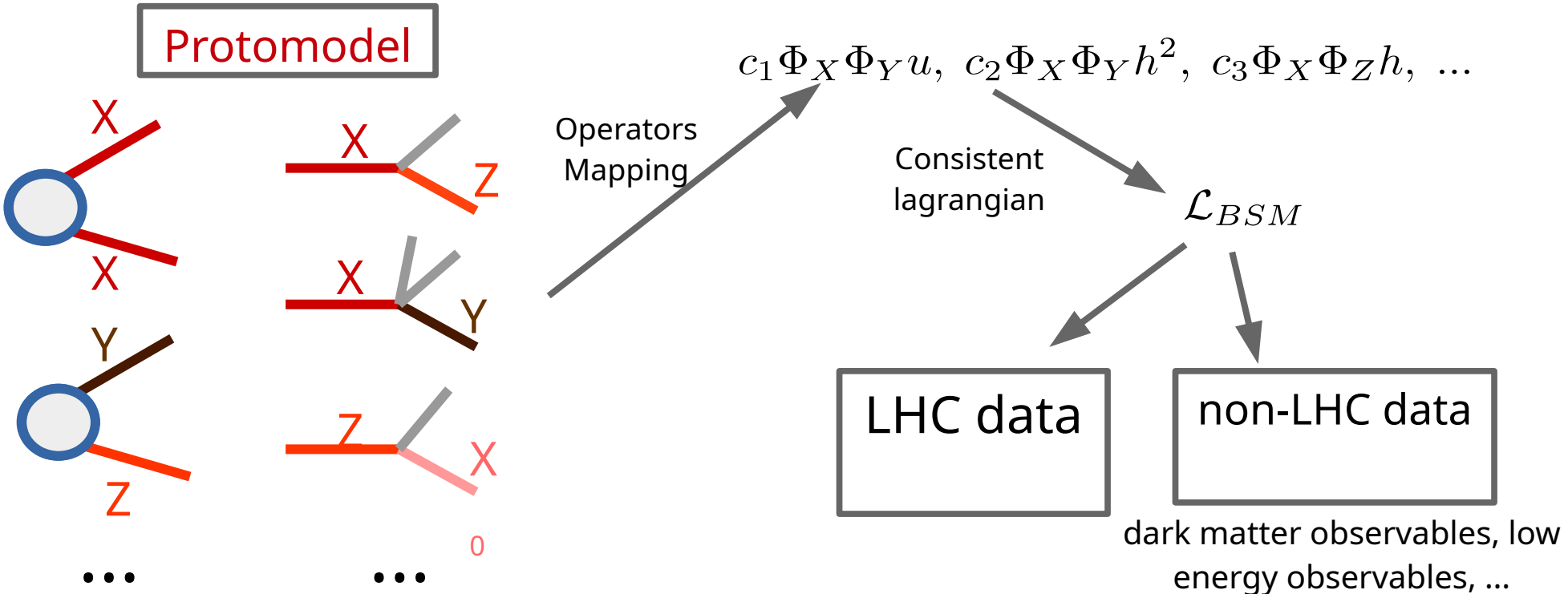
Summary

Well.

ProtoModels → Next Standard Model

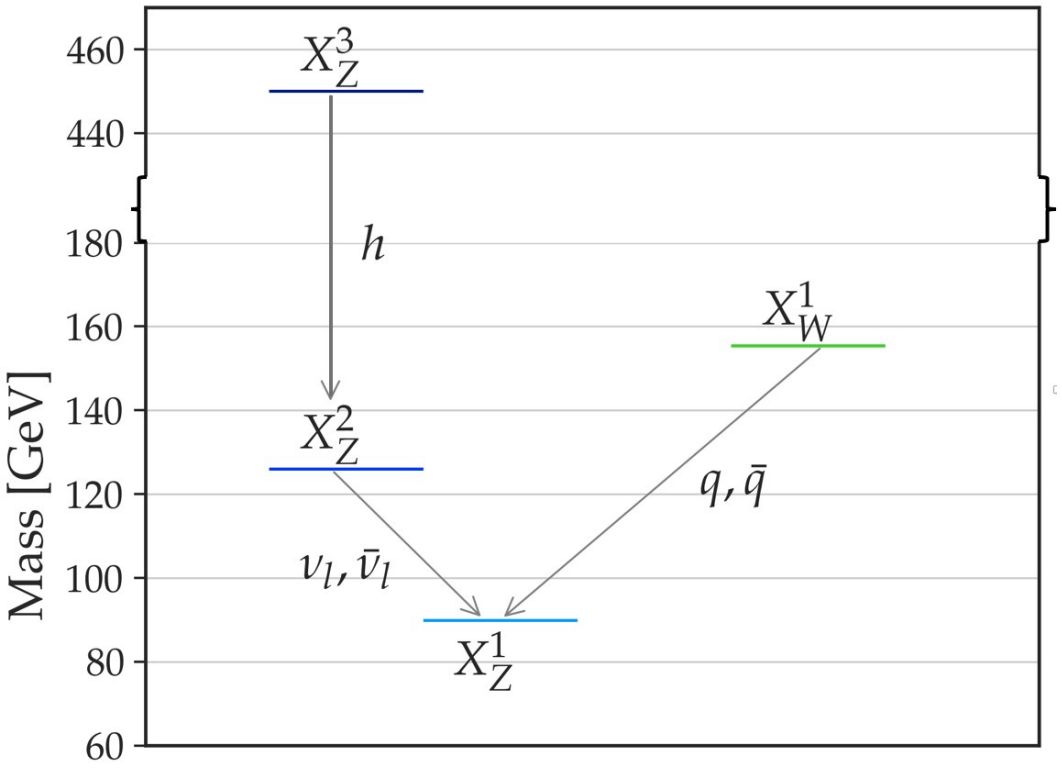


Work from protomodels to UV complete theories has begun
(John Gargalionis, PostDoc in Valencia) – lot's of combinatorics!



Here is such a prototype model, a “protomodel”

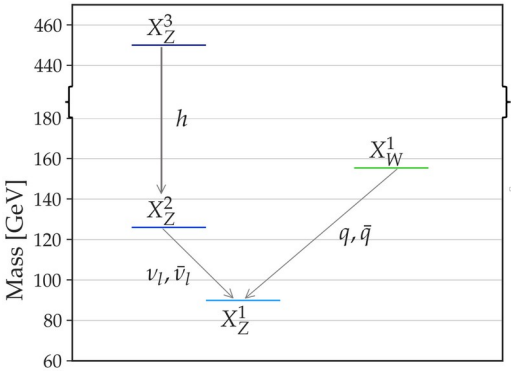
(We introduced this notion of a “protomodel”. It’s an incomplete BSM model, driven by data, not by theory considerations. LHC production cross sections are free parameters, BSM particle spins are left unspecified)



X^i_Z = “neutralino-like”

X^i_W = “chargino-like”

These were typical masses for wino-like neutralinos, for higgsino-like neutralinos: 500 GeV



Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
CMS-EXO-20-004	comb	-	TChiISR, TChiZISRqq	2e+02 fb	1e+02 fb	4.6 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-EXOT-2018-06	em	EM10	TChiISR, TChiZISRqq	413	359.00 +/- 10.00	2.5 σ	X^1_W, X^1_Z, X^2_Z
CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.1 σ	X^3_Z, X^2_Z

*) It contextualizes three of the biggest excesses we are seeing in the databases, that are considered mutually independent.

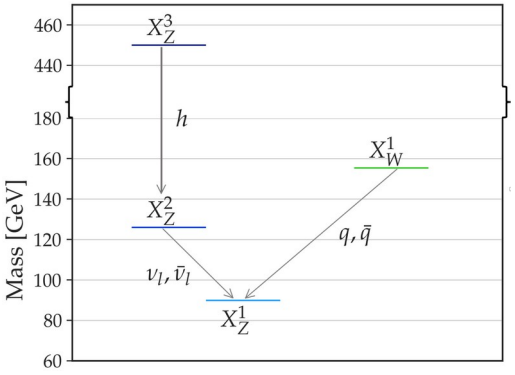
*) we can thus approximately combine the likelihoods of these three analyses – by multiplying them.

*) Another result exists, but is correlated, so we cannot add it in that fashion (but it would also exhibit an excess)

$$T_L := 2 \ln \left(\frac{L^c_{\mathrm{BSM}}(\hat{\mu}) \cdot \pi_{\mathrm{BSM}}}{L^c_{\mathrm{SM}} \cdot \pi_{\mathrm{SM}}} \right) = 33.69 = 5.8^2$$

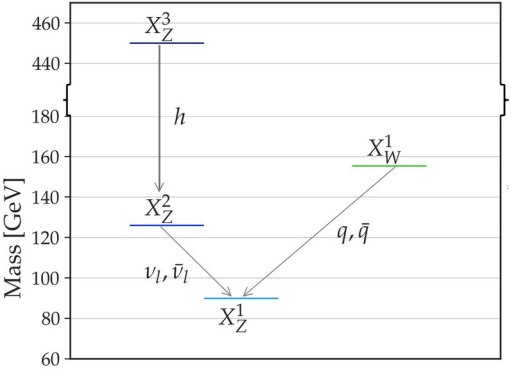
"local 5.8 sigma" (if ndf=1) – with a GIGANTIC look elsewhere effect!

But let's focus on the simpler protomodel



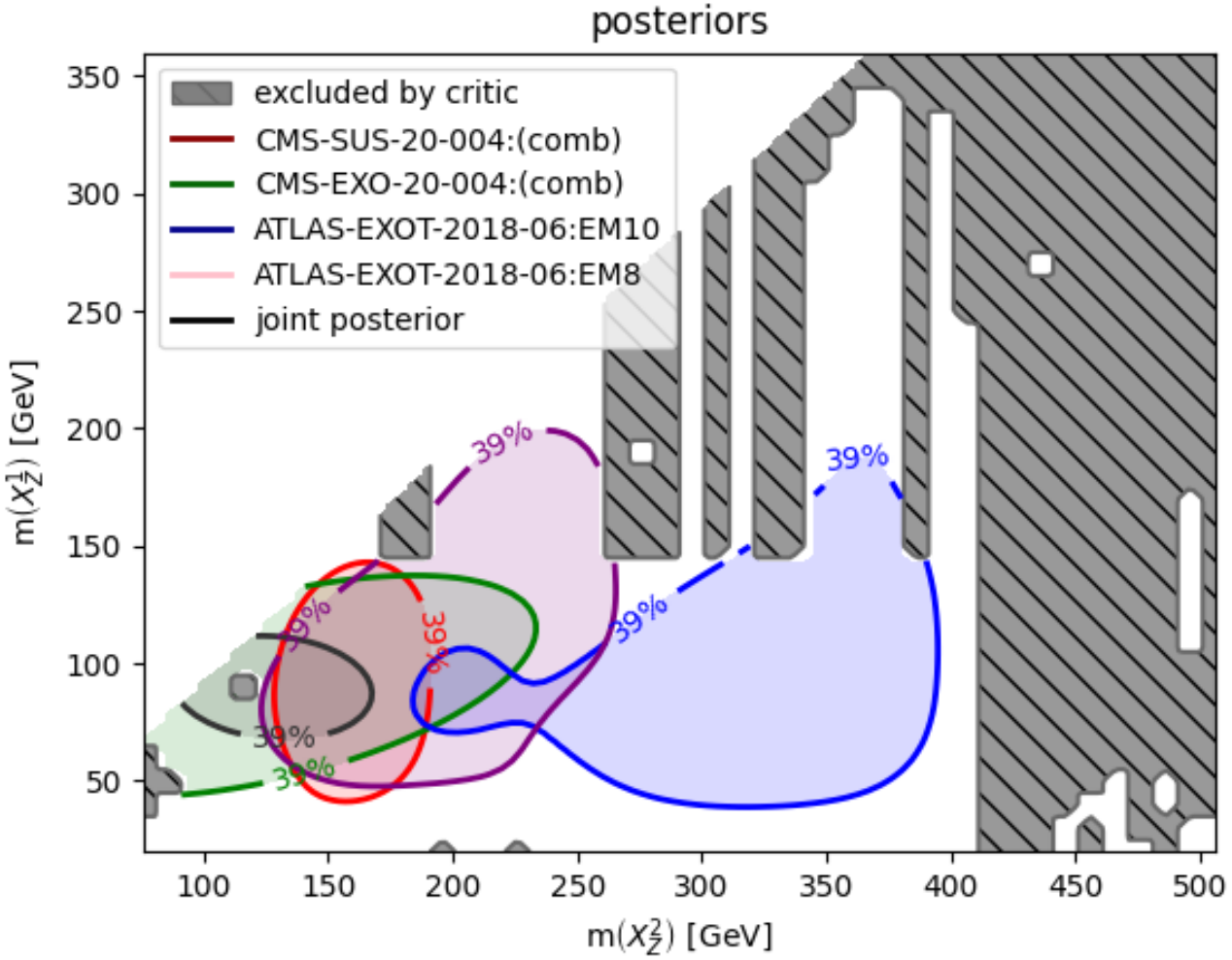
Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
CMS-EXO-20-004	comb	-	TChiISR, TChiZISRqq	2e+02 fb	1e+02 fb	4.6 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-EXOT-2018-06	em	EM10	TChiISR, TChiZISRqq	413	359.00 +/- 10.00	2.5 σ	X^1_W, X^1_Z, X^2_Z
CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.1 σ	X^3_Z, X^2_Z

Do we consider these excesses to be mutually compatible?

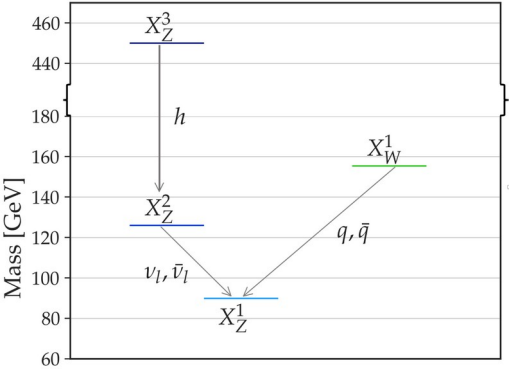


Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
CMS-EXO-20-004	comb	-	TChiISR, TChiZISRqq	2e+02 fb	1e+02 fb	4.6 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-EXOT-2018-06	em	EM10	TChiISR, TChiZISRqq	413	359.00 +/- 10.00	2.5 σ	X^1_W, X^1_Z, X^2_Z
CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.1 σ	X^3_Z, X^2_Z

Combining Likelihoods

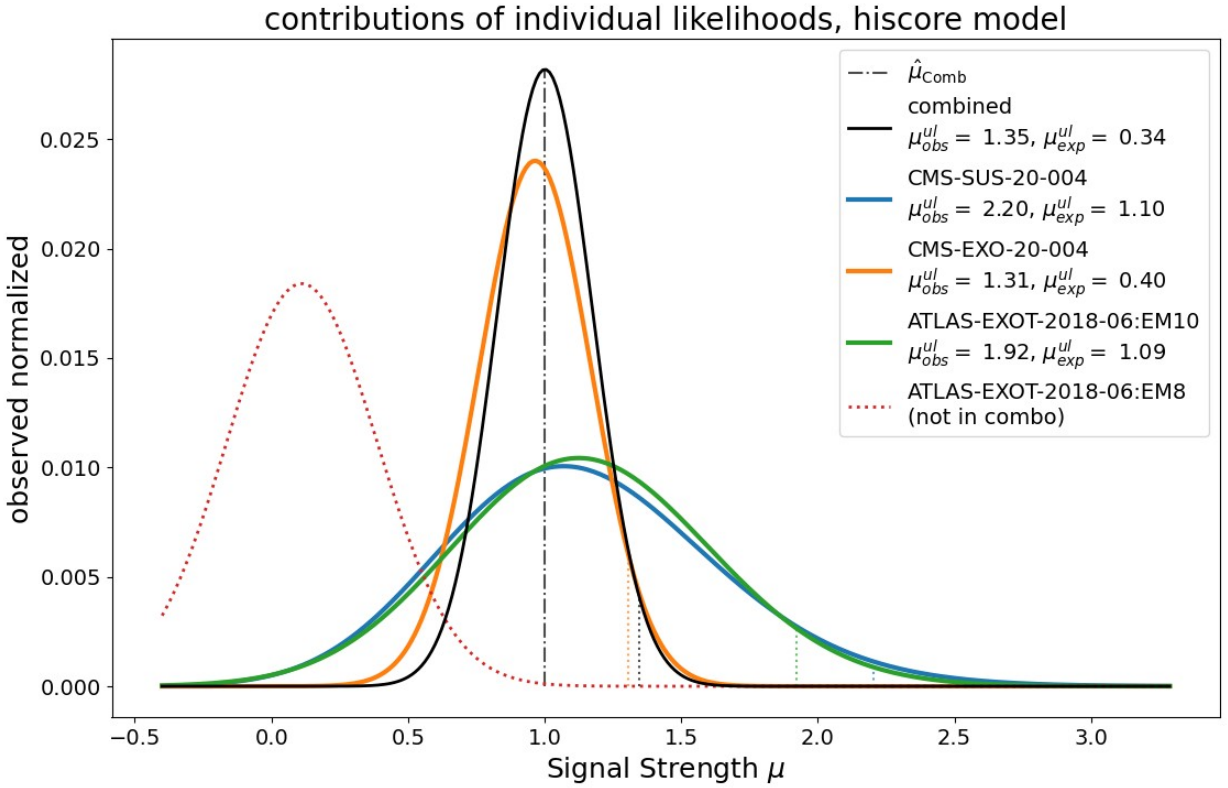


Likelihoods as functions of particle masses



Analysis Name	Type	Dataset	Topos	Observed	Expected	Approx σ	Particles
CMS-EXO-20-004	comb	-	TChiISR, TChiZISRqq	2e+02 fb	1e+02 fb	4.6 σ	X^1_W, X^1_Z, X^2_Z
ATLAS-EXOT-2018-06	em	EM10	TChiISR, TChiZISRqq	413	359.00 +/- 10.00	2.5 σ	X^1_W, X^1_Z, X^2_Z
CMS-SUS-20-004	comb	-	TChiHH	0.3 fb	0.2 fb	2.1 σ	X^3_Z, X^2_Z

Combining Likelihoods



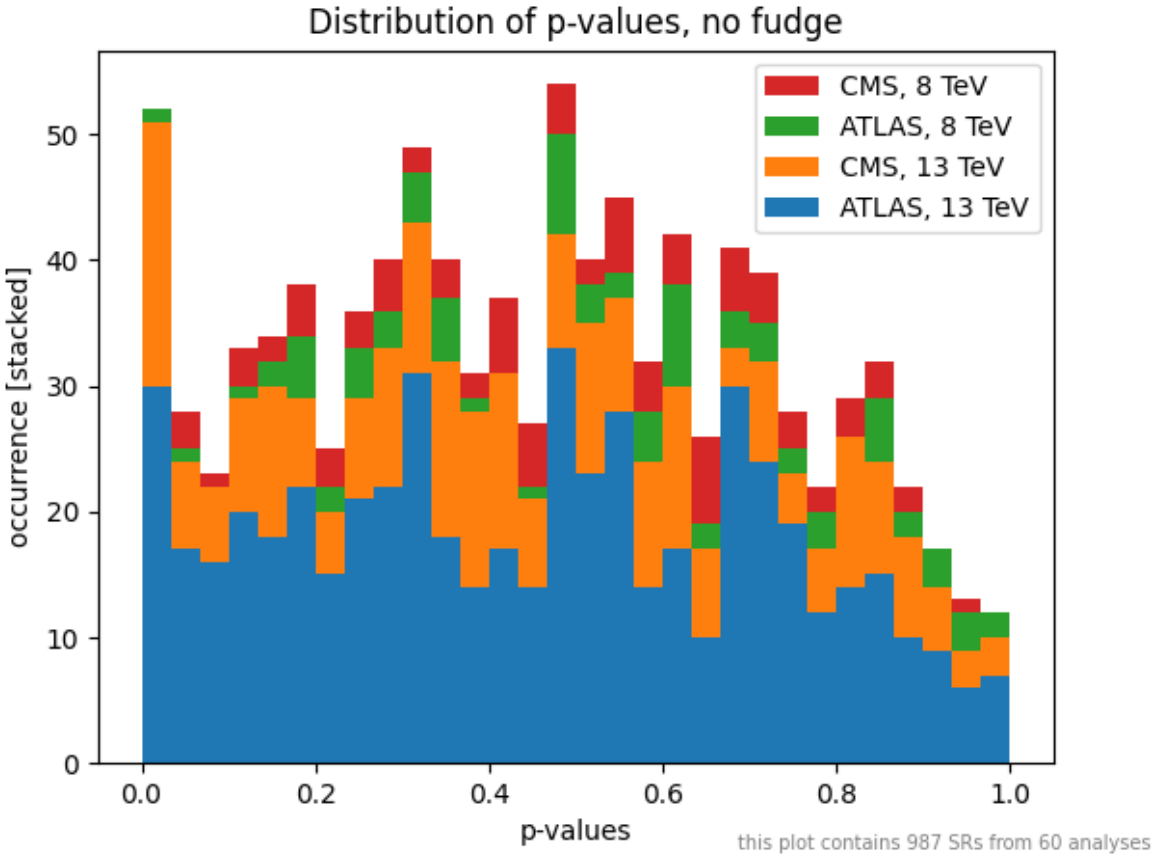
Likelihoods as functions of an overall signal strength multiplier

Taking this model with mu as the only free parameter, we could formulate a likelihood ratio test:

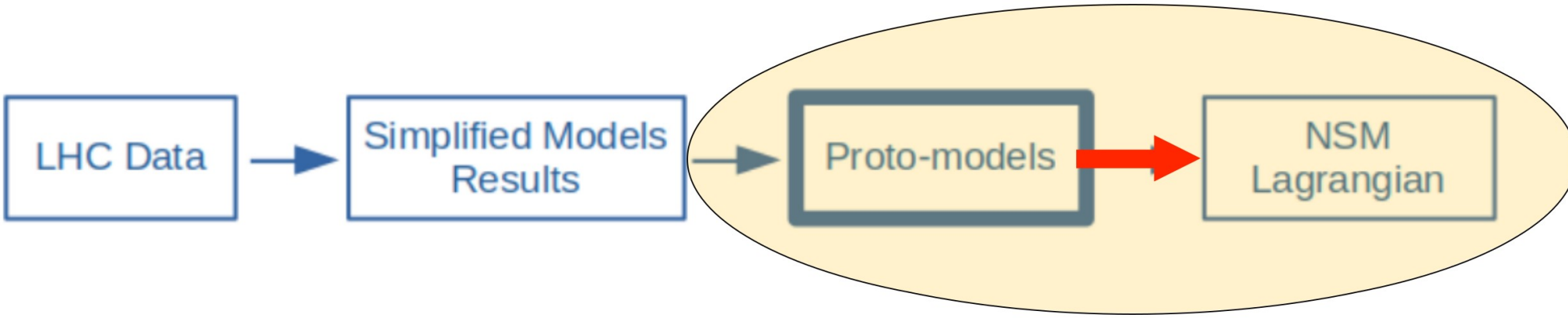
$$T_L := 2 \ln \left(\frac{L^c_{\text{BSM}}(\hat{\mu}) \cdot \pi_{\text{BSM}}}{L^c_{\text{SM}} \cdot \pi_{\text{SM}}} \right) = 33.69 = 5.8^2$$

Now that we have a protomodel, what about the meta-statistics plots that I showed you earlier? We have a new hypothesis, so we can compute the p-values for that!

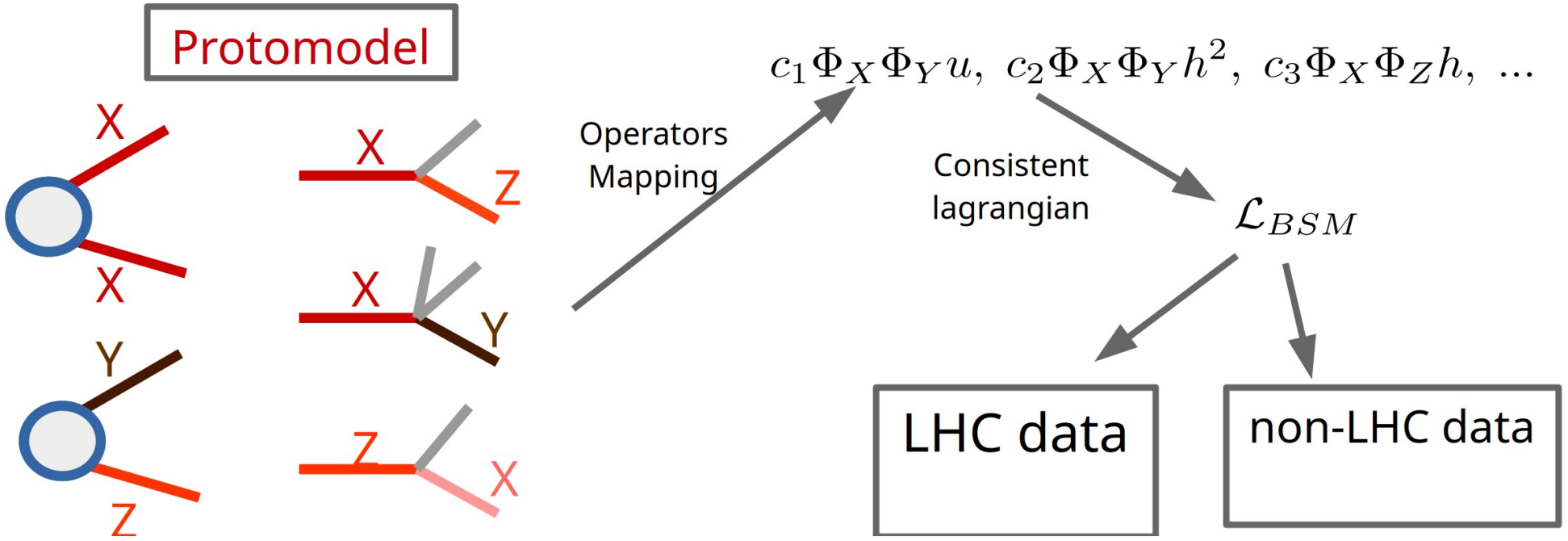
For the SM hypothesis:



NSM Lagrangian from the proto-models, work in progress

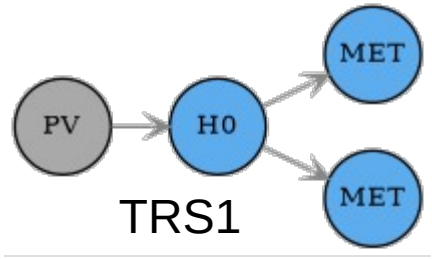


Work in progress. Lots of combinatorics:
we cannot differentiate between light quark flavors. Observed particles might be degenerate.
From particles to fields.
Singlets versus doublets versus multiplets. From broken to unbroken Lagrangians.

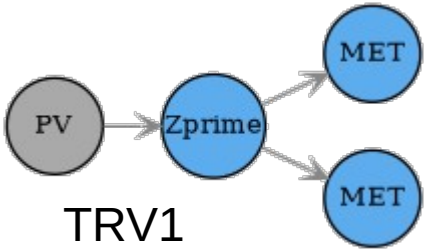


Naively: (*without* assuming conservatism) we would expect
<= 960 * Φ(2.40) ≈ 8 SRs.
We have 28 SRs with Z > 2.40, 4 SRs with Z < - 2.40

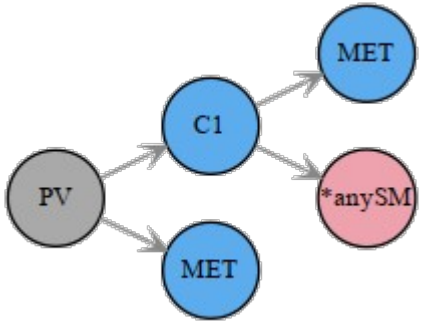
List Of SRs with Z > 2.40



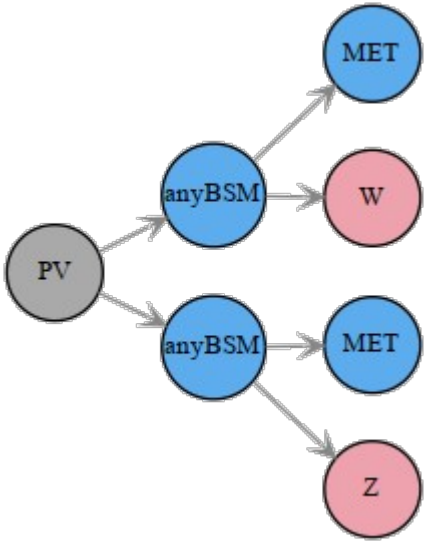
TRS1



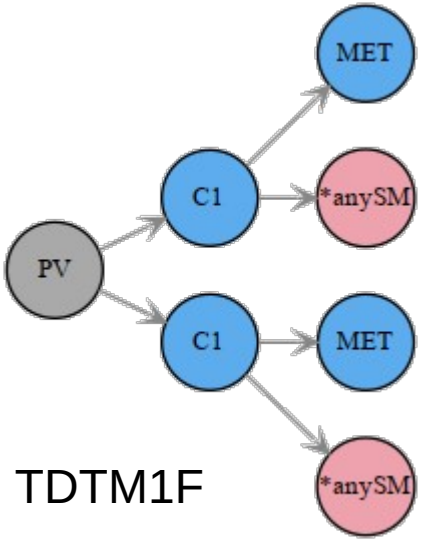
TRV1



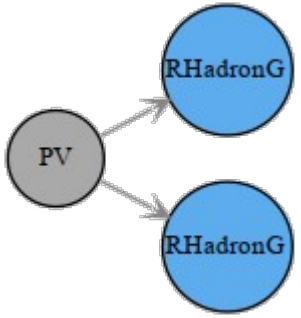
TDTM2F



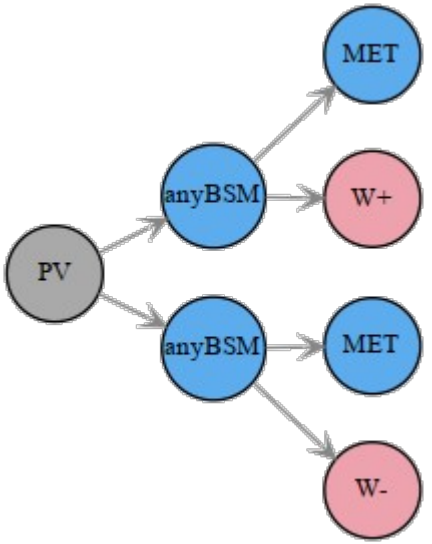
TChiWZ(off)



TDTM1F



Z	analysis	SR	topo	obsN	expected
4.47	CMS-EXO-20-004	SR_2018_8	TChiISR ...	8529	7720.98±152.03
4.31	CMS-EXO-20-004	SR_2018_4	TChiISR ...	35833	33037.13±618.12
4.11	CMS-EXO-20-004	SR_2018_6	TChiISR ...	14266	13209.01±250.28
3.85	CMS-EXO-20-004	SR_2018_9	TChiISR ...	5406	4914.64±101.04
3.80	ATLAS-SUSY-2018-42	SR1400_High_long	T1DispT ...	7	0.70±0.40
3.78	CMS-EXO-20-004	SR_2018_1	TChiISR ...	191829	178470.12±3615.00
3.72	CMS-EXO-20-004	SR_2018_2	TChiISR ...	104522	97428.42±1896.81
3.63	CMS-EXO-20-004	SR_2018_3	TChiISR ...	59398	55592.19±1059.45
3.46	ATLAS-SUSY-2018-42	SR1500_High_long	TDTM1FT ...	6	0.60±0.40
3.35	CMS-EXO-20-004	SR_2018_11	TChiISR ...	2671	2428.39±55.08
3.30	ATLAS-SUSY-2018-16-hino	SR_ewk_2l_med	TChiWWo ...	170	112.40±12.60
3.21	ATLAS-SUSY-2018-42	SR1300_High_long	TDTM1FT ...	6	0.80±0.40
3.21	ATLAS-SUSY-2018-16	SR_ewk_2l_ee_high_d	TChiWZoff	16	4.80±1.60
3.17	CMS-EXO-20-004	SR_2018_12	TChiISR ...	1613	1441.83±35.64
3.17	ATLAS-SUSY-2018-42	SR1600_High_long	T1DispT ...	5	0.54±0.35
3.01	ATLAS-SUSY-2018-42	SR1200_High_long	T1DispT ...	6	0.90±0.50
3.01	CMS-EXO-20-004	SR_2018_7	TChiISR ...	11730	10991.22±213.74
2.93	ATLAS-SUSY-2017-03	SR3l_ISR	TChiWZ	12	3.90±1.00
2.83	ATLAS-SUSY-2018-42	SR1800_High_long	T1DispTRHadGM1	4	0.44±0.32
2.75	CMS-EXO-20-004	SR_2018_5	TChiISR ...	21854	20714.72±391.42
2.56	ATLAS-SUSY-2019-09	SRlow_0Jd	TChiWZoff	77	54.00±4.00
2.51	CMS-SUS-20-004	SR11	T5HHTChiHH	4	0.06±0.11
2.51	CMS-EXO-20-004	SR_2018_10	TChiISR ...	3285	3065.69±66.79
2.49	ATLAS-SUSY-2018-42	SR2400_High_long	T1DispTRHadGM1	3	0.29±0.28
2.47	ATLAS-EXOT-2018-06	EM10	TChiISR ...	413	359.00±10.00
2.45	ATLAS-SUSY-2018-42	SR2600_High_long	T1DispTRHadGM1	3	0.25±0.31
2.44	ATLAS-SUSY-2018-42	SR2000_High_long	T1DispTRHadGM1	3	0.36±0.25
2.42	ATLAS-SUSY-2018-42	SR2200_High_long	T1DispTRHadGM1	3	0.33±0.28



*) CMS-EXO-20-004 is yes overrepresented
(simplified likelihood fit possible culprit?)
*) “small mass gap + MET” topologies are dominant